CHAPTER 6

EQUILIBRIUM

CHAPTER ESSENCE AT A GLANCE:
Covers dynamic chemical equilibrium, Le Chatelier's principle, acid-base theories, pH calculations, buffer systems, and solubility product.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Dynamic Equilibrium & Law of Mass Action:** Forward and reverse reaction rates equalize at equilibrium; Kp = Kc (RT)^{D}elta n$_{g}$.
- **Le Chatelier's Principle:** A system at equilibrium disturbed by changes in T, P, or concentration shifts to counteract the perturbation.
- **Acid-Base Concepts & pH Scale:** Bronsted-Lowry involves proton transfer; Lewis involves electron pairs; pH = -log10[H+].
- **Buffer Action & Common Ion Effect:** Buffer solutions resist pH change; common ions suppress weak electrolyte dissociation.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$K_p = K_c(RT)^{\Delta n_g}$$

Relation Between Partial Pressure and Molar Equilibrium Constants

$$\mathrm{pH} = -\log_{10}[\mathrm{H}^+] \quad \text{and} \quad \mathrm{pH} + \mathrm{pOH} = 14 \text{ at } 298 \text{ K}$$

pH Scale and Ionic Product of Water

$$\mathrm{pH} = \mathrm{p}K_a + \log\left(\frac{[\mathrm{Salt}]}{[\mathrm{Acid}]}\right)$$

Henderson-Hasselbalch Equation for Acidic Buffer

$$K_{\mathrm{sp}} = [M^{y+}]^x[A^{x-}]^y$$

Solubility Product Constant for Sparingly Soluble Salts

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** Adding a catalyst accelerates forward and reverse reactions equally and has ZERO effect on Kc or equilibrium position.
- **Alert:** Concentrations of pure solids and pure liquids are constant and omitted from equilibrium constant expressions.
- **Alert:** In Haber's ammonia synthesis, increasing pressure shifts equilibrium forward because gaseous moles decrease from 4 to 2.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State Le Chatelier's Principle. What is the effect of adding a catalyst on the position of chemical equilibrium? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Le Chatelier's Principle:** If a system at chemical equilibrium is subjected to a disturbance in temperature, pressure, or concentration, the equilibrium shifts in a direction that tends to counteract or undo the effect of that change. **[1 Mark]**

• **Effect of Catalyst:** A catalyst increases the rates of both forward and reverse reactions to the identical extent by lowering activation energy. Hence, a catalyst has ZERO effect on the position of equilibrium or equilibrium constant. **[1 Mark]**

• **Key Exam Result:** Catalyst speeds attainment of equilibrium without altering equilibrium composition.

2. Define conjugate acid-base pairs according to Bronsted-Lowry concept. Write the conjugate bases of HCO3- and H2O. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** An acid-base pair that differs by only one single proton (H+) is called a conjugate acid-base pair. Acid -> H+ + Conjugate Base. **[1 Mark]**

• **Conjugate Bases:** Removing H+ from HCO3- yields CO3(2-) (Carbonate ion). Removing H+ from H2O yields OH- (Hydroxide ion). **[1 Mark]**

• **Key Exam Result:** Conjugate base of HCO3- is CO3(2-); Conjugate base of H2O is OH-.

3. Derive the mathematical relationship Kp = Kc (R T)^Δn for a general reversible gaseous reaction. Under what condition is Kp equal to Kc? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Ideal Gas Partial Pressure):** For a reaction aA + bB ≤ > cC + dD, partial pressure $p_i = (n_i$ divided by V) R T = [i] R T. **[1 Mark]**

• **Step 2 (Substitution in Kp):** Kp = $(p_C^c \cdot p_D^d)$ divided by $(p_A^a \cdot p_B^b)$ = $([C]^c [D]^d$ divided by $[A]^a [B]^b) \cdot$ (R T)^((c+d)-(a+b)). **[1 Mark]**

• **Step 3 (Final Expression):** Kp = Kc (R T)^Δ n_g, where Δ n_g = (c + d) - (a + b). If Δ n_g = 0 (equal gaseous moles on both sides, e.g., H2 + I2 ≤ > 2HI), then $(R T)^0 = 1$, so Kp = Kc. **[1 Mark]**

• **Key Exam Result:** Kp = Kc (RT)^Δn; Kp equals Kc when Δ n_g = 0.

$$K_p = K_c(RT)^{\Delta n_g} \quad (\Delta n_g = 0 \rightarrow K_p = K_c)$$

Equilibrium Constant Relation

4. Apply Le Chatelier's principle to Haber's ammonia synthesis: N2 (g) + 3H2 (g) ≤ > 2NH3 (g), ΔH = -92.4 kJ/mol. State the optimal conditions of temperature, pressure, and catalyst for maximum ammonia yield. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Effect of Pressure:** Forward reaction involves decrease in gaseous moles (4 mol -> 2 mol). Increasing pressure shifts equilibrium forward. Optimum pressure is 200 atm. **[1 Mark]**

• **Effect of Temperature:** Forward reaction is exothermic (ΔH = -92.4 kJ/mol). Low temperature favours forward yield, but too low reduces reaction rate. Optimum temperature is ~700 K. **[1 Mark]**

• **Catalyst & Concentration:** Iron oxide catalyst with K2O and Al2O3 promoter speeds attainment of equilibrium; continuous removal of NH3 shifts equilibrium forward. **[1 Mark]**

• **Key Exam Result:** Optimal conditions: ~700 K, 200 atm pressure, iron oxide catalyst.

5. What is the common ion effect? Explain its important practical application in the qualitative analysis of Group II basic radicals. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** The suppression of the ionization and dissociation of a weak electrolyte by the addition of a strong electrolyte containing a common ion. **[1 Mark]**

• **Group II Qualitative Analysis:** Group II cations (Cu2+, Pb2+) are precipitated as sulfides using H2S gas in the presence of dilute HCl. **[1 Mark]**

• **Mechanism:** HCl is a strong acid that provides high [H+] ions, suppressing H2S ionization. [S2-] is reduced so that only Group II cations (having very low Ksp) exceed their solubility product, preventing higher group cations from precipitating. **[1 Mark]**

• **Key Exam Result:** Common ion effect controls sulfide concentration for selective cation precipitation.

6. Calculate the pH of: (a) 0.001 M HCl solution, (b) 0.05 M Ba(OH)2 solution at 298 K. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Part (a) 0.001 M HCl:** HCl is a strong acid, so [H+] = 0.001 M = 10^{-3} M. pH = -log10[H+] = -log10(10^{-3}) = 3.0. **[1.5 Marks]**

• **Part (b) 0.05 M Ba(OH)2:** Ba(OH)2 dissociates into Ba2+ + 2OH-. Therefore, [OH-] = 2 × 0.05 M = 0.1 M = 10^{-1} M. **[0.5 Mark]**

• **Part (b) pH Calculation:** pOH = -log10[OH-] = -log10(10^{-1}) = 1.0. At 298 K, pH = 14 - pOH = 14 - 1.0 = 13.0. **[1 Mark]**

• **Key Exam Result:** pH of 0.001 M HCl = 3.0; pH of 0.05 M Ba(OH)2 = 13.0.

$$\mathrm{pH(HCl)} = -\log(10^{-3}) = 3.0, \quad \mathrm{pH(Ba(OH)_2)} = 14 - [-\log(0.1)] = 13.0$$

Acidic and Basic Solution pH Calculation

7. Define a buffer solution. Derive the Henderson-Hasselbalch equation for the pH of an acidic buffer mixture. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** A solution which resists any change in its pH value upon the addition of small amounts of strong acid, strong base, or upon dilution. **[1 Mark]**

• **Derivation:** For weak acid HA <= > H+ + A-, Ka = ([H+][A-]) divided by [HA]. Rearranging: [H+] = Ka · ([HA] divided by [A-]). **[1 Mark]**

• **Logarithmic Form:** Taking negative logarithms: -log[H+] = -log Ka - log([HA] divided by [A-]). Hence, pH = pKa + log([A-] divided by [HA]) = pKa + log([Salt] divided by [Acid]). **[1 Mark]**

• **Key Exam Result:** Henderson-Hasselbalch equation: pH = pKa + log($\frac{[Salt]}{[Acid]}$).

$$\mathrm{pH} = \mathrm{p}K_a + \log\left(\frac{[\mathrm{Salt}]}{[\mathrm{Acid}]}\right)$$

Henderson-Hasselbalch Buffer Formulation